**MGMT 6600 Term Project Report**

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Level: 6000

**Abstract**: This analysis aims to answer how changes in monetary policy affect equity returns and volatility over time. A separate, personal goal of the project involves strengthening my financial analytics skills before beginning my professional career as an Underwriting Analyst following graduation. The research loosely builds on a prior MBA project in which macroeconomic data was analyzed to determine whether a specific shock event, the release of ChatGPT in November 2022, influenced labor productivity across various industries.

Stock markets tend to reflect fluctuations in the macroeconomy and business cycle. The goal of this project is to examine meaningful relationships among different macroeconomic variables and with the greater equity market. Specifically, the project aims to identify how asset returns and volatilities are impacted by major shock events and changes in monetary policy (e.g. interest rates), and how these adjustments affect asset correlations and buyer behavior. The preliminary hypothesis suggests that policy changes *do* impact asset returns and volatility in a statistically significant way.

# I. Data Description

The datasets were chosen with intent to track macroenvironment changes over time and to predict equity returns and volatility. The first dataset, sourced from Kaggle, contains daily Yahoo Finance metrics on many corporate stocks (n = 491) over a five-year period (November 29, 2018 - November 29, 2023). The dataset is robust (n > 600,000) and explores various asset pricing measurements (high/low, open/close), volume, and corporate actions. Capturing two recent shock events (COVID-19 and AI tech boom), this time period provides a space to directly assess the impact on equities.

The second dataset, sourced from the Federal Reserve of Economic Data (FRED), contains *specific* macroenvironmental parameters chosen to measure existing relationships and the impact of policy changes on consumer behavior. Covering a wider period from 1962-2024, this dataset enables both a historical analysis as well as a five-year overlapping period with the equity dataset. For this dataset, the data dictionary is as follows:

| Data Dictionary | | |
|---|---|---|
| **Variable Name** | **Variable Description** | **Last Updated (FRED)** |
| FPCPITOTLZGUSA | Inflation, consumer prices for the United States, Percent, Annual, Not Seasonally Adjusted | 4/16/2025 |
| GDPA | Gross Domestic Product, Billions of Dollars, Annual, Not Seasonally Adjusted | 9/25/2025 |
| GFDEBTN | Federal Debt: Total Public Debt, Millions of Dollars, Quarterly, Not Seasonally Adjusted | 1/22/2026 |
| M2NS | M2, Billions of Dollars, Monthly, Not Seasonally Adjusted | 1/27/2026 |
| MEHOINUSA672N | Real Median Household Income in the United States, 2024 C-CPI-U Dollars, Annual, Not Seasonally Adjusted | 9/9/2025 |
| RIFLGFCY10NA | Market Yield on U.S. Treasury Securities at 10-Year Constant Maturity, Quoted on an Investment Basis, Percent, Annual, Not Seasonally Adjusted | 1/2/2026 |
| RIFSPFFNA | Federal Funds Effective Rate, Percent, Annual, Not Seasonally Adjusted | 1/2/2026 |
| UNRATENSA | Unemployment Rate, Percent, Monthly, Not Seasonally Adjusted | 1/9/2026 |

These metrics were chosen because they capture different aspects of monetary policy and consumer behavior. While the federal funds rate captures direct actions taken by the Federal Reserve, other metrics like GDP, real median household income, and M2 reflect the economy's growth over time. Finally, the market yield on 10-year Treasuries was included to provide an approximation of the **risk-free rate**, which will be useful when comparing returns and volatilities among riskier assets.

# II. Exploratory Analysis

## 1. *Cleaning and Preparing Datasets*

The first step involved organizing the macro data to use a standardized annual time scale across all variables. Although each metric initially covered slightly different time periods, there is an overlap from **1984-2024**, providing 40 years of data to analyze. For metrics using a monthly time scale, the first observation from each year was extracted to align with an annual frequency.

To produce summary statistics of the equity variables, a random subset of stock tickers was selected to avoid computational overload. This was performed in R and a subset of 40 tickers was selected, resulting in approximately 48,000 observations across the five-year period. By reducing the number of tickers analyzed to less than 10% of the original dataset, it is possible that the sample may not truly be representative. A random sampling technique was chosen to eliminate selection bias.

Despite the reduced sample size, limitations still occurred while performing exploratory analysis. As a result, the daily equity data for each sample ticker were aggregated to a monthly frequency. For volume, dividends, and stock splits, this equaled

the sum across all days in a given month for each ticker. The high and low prices were calculated as the maximum and minimum of each ticker's high and low prices for each month, respectively. Monthly open and close prices were calculated as the opening price on the first of the month and the closing price on the final day. After aggregating monthly, there were approximately 2,318 observations across 40 different stocks. Using the monthly data, additional metrics were computed for each company. **Monthly returns** were calculated based on changes in closing prices, and **market volatility** was measured using a rolling 12-month standard deviation of monthly returns.

Finally, the **monthly** equity metrics were aggregated to **yearly** observations in order to align with the macroeconomic dataset. Returns and volatility were averaged across the months within each year, while total volume, dividends, and stock splits were summed. Yearly high and low prices were calculated as the maximum and minimum monthly values observed during each year. This final aggregation produced a **combined dataset containing yearly macroeconomic indicators alongside measures of equity market performance** from **2018–2023**.

2. *EDA: FRED (Macro) Dataset*

Summary statistics for each macro variable were computed in Excel across the sample period (Table 1). GDP exhibits the largest magnitude, with an approximate average of $6.08 trillion and a wide range spanning from $57 billion to nearly $29.3 trillion. This variation reflects the wide time horizon of this variable's measurement (1929-2024) and the *significant* growth of the U.S. economy over the past several decades. Other measures of monetary expansion such as M2 and federal debt demonstrate similar large ranges, indicating that these variables have increased substantially over time.

| GDP, $ Billions | |
|---|---|
| **Mean** | 6080.273469 |
| Standard Error | 776.4940953 |
| Median | 1977.619 |
| **Standard Deviation** | 7608.057287 |
| Sample Variance | 57882535.69 |
| Kurtosis | 0.716842582 |
| Skewness | 1.295815091 |
| Range | 29240.859 |
| Minimum | 57.154 |
| Maximum | 29298.013 |
| Count | 96 |

| Fed Funds Rate, % | |
|---|---|
| **Mean** | 4.629859 |
| Standard Error | 0.412156 |
| Median | 4.22 |
| **Standard Deviation** | 3.472887 |
| Sample Variance | 12.06094 |
| Kurtosis | 1.053724 |
| Skewness | 0.933052 |
| Range | 16.31 |
| Minimum | 0.08 |
| Maximum | 16.39 |
| Count | 71 |

| Inflation, % | |
|---|---|
| **Mean** | 3.757704479 |
| Standard Error | 0.340228477 |
| Median | 3.015075377 |
| **Standard Deviation** | 2.743009678 |
| Sample Variance | 7.524102094 |
| Kurtosis | 2.786326333 |
| Skewness | 1.613038298 |
| Range | 13.90474824 |
| Minimum | -0.355546266 |
| Maximum | 13.54920197 |
| Count | 65 |

| Mkt Yield on 10yr Treasuries, % | |
|---|---|
| **Mean** | 5.82328125 |
| Standard Error | 0.365828998 |
| Median | 5.45 |
| **Standard Deviation** | 2.926631985 |
| Sample Variance | 8.565174777 |
| Kurtosis | 0.281397216 |
| Skewness | 0.717662901 |
| Range | 13.03 |
| Minimum | 0.89 |
| Maximum | 13.92 |
| Count | 64 |

| Real Median Household Income, $ | |
|---|---|
| **Mean** | 70423.90244 |
| Standard Error | 976.7196766 |
| Median | 70040 |
| **Standard Deviation** | 6254.057434 |
| Sample Variance | 39113234.39 |
| Kurtosis | -0.283728126 |
| Skewness | 0.663712033 |
| Range | 23310 |
| Minimum | 60420 |
| Maximum | 83730 |
| Count | 41 |

| Unemployment Rate, % | |
|---|---|
| **Mean** | 6.238462 |
| Standard Error | 0.196551 |
| Median | 6.15 |
| **Standard Deviation** | 1.735888 |
| Sample Variance | 3.013307 |
| Kurtosis | 0.071464 |
| Skewness | 0.62214 |
| Range | 8 |
| Minimum | 3.4 |
| Maximum | 11.4 |
| Count | 78 |

| Federal Debt, $ Millions | |
|---|---|
| **Mean** | 8457625.733 |
| Standard Error | 1252784.074 |
| Median | 4990951 |
| **Standard Deviation** | 9704023.71 |
| Sample Variance | 9.41681E+13 |
| Kurtosis | 1.00003822 |
| Skewness | 1.380058911 |
| Range | 35893311 |
| Minimum | 320999 |
| Maximum | 36214310 |
| Count | 60 |

| M2 Money Supply, $ Billions | |
|---|---|
| **Mean** | 5526.104478 |
| Standard Error | 728.0002738 |
| Median | 3383.2 |
| **Standard Deviation** | 5958.939059 |
| Sample Variance | 35508954.71 |
| Kurtosis | 1.205892839 |
| Skewness | 1.425348164 |
| Range | 21351.6 |
| Minimum | 289.8 |
| Maximum | 21641.4 |
| Count | 67 |

Table 1: Summary Statistics for Macroeconomic Variables

Interest rate and inflation measures show more moderate variation. The federal funds rate averages 4.63%, while the market yield on 10-year Treasuries averages about 5.82%. The positive skewness exhibited by both suggest few high interest rate spikes over the sample period. Inflation averages 3.76% with *significant* variability, ranging from slight deflation (2009) to over 13%. The unemployment rate averages 6.24% with relatively lower variability. Finally, real median household income averages

$70,424 over its sample period (1984-2024) with modest variation, indicating more gradual changes in household purchasing power over time relative to other growth indicators such as GDP or M2.

Figure 1 illustrates the behavior of several macro indicators, including the federal funds rate, inflation, the market yield on 10-year Treasuries, and the unemployment rate. The chart highlights the significant volatility in interest rates and inflation during the late 1970s and early 1980s, when both variables historically peaked as monetary policy tightened in response to rapidly rising inflation. Following this period, both inflation and interest rates declined sharply and remained moderate throughout the early 2000s. Around the 2008 financial crisis, a sharp decline in the federal funds rate is visible and is once again during the COVID-19 period, reflecting reactive monetary policy actions. The unemployment rate increases during recessionary periods, particularly during the early 1980s and again following the 2008 crisis, demonstrating the relationship between monetary policy and labor market conditions.

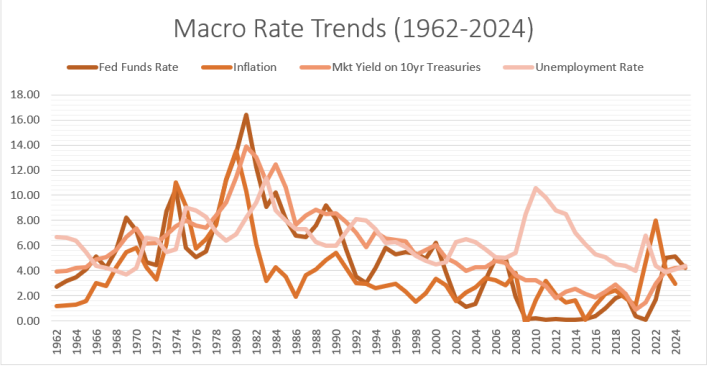



Figure 1: Macroeconomic Rate Trends (1962-2024)

Figure 2 plots long-term trends in GDP, real median household income, the M2 money supply, and federal public debt, providing insight into economic growth over time. All variables display upward trends, reflecting long-term economic expansion and monetary growth. Federal debt shows particularly rapid growth following the 2008 financial crisis and again during the COVID-19 pandemic. Real median household income experiences periods of modest growth followed by temporary declines during economic downturns.

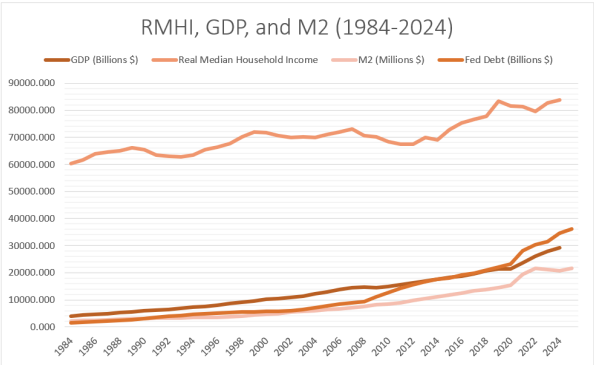



Figure 2: Real Median Household Income, GDP, and M2 Money Supply (1984-2024)

Figure 3 plots the federal funds rate (x) with total federal debt (y). The negative relationship suggests higher levels of federal debt tend to occur during periods of lower interest rates. While this relationship does not imply causation, $R^2$ value of 32% suggests that interest rates and federal borrowing levels move inversely over time.

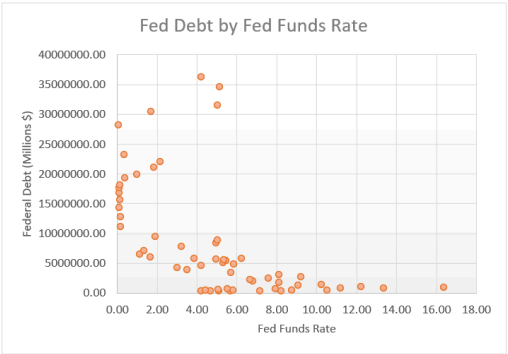
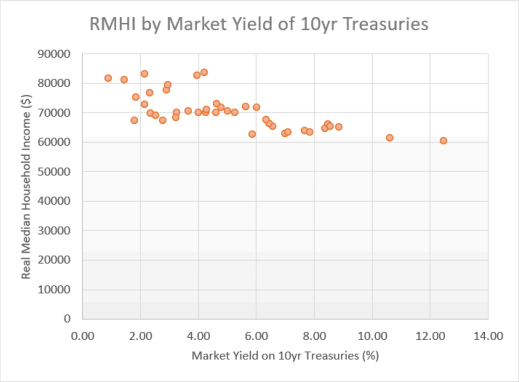



Figure 3: Federal Debt and Federal Funds Rate

Figures 4 and 5 examine the relationship between real median household income (RMHI) and two indicators: the market yield on 10-year Treasuries, and the unemployment rate, respectively. The first scatterplot shows a negative relationship, with the trendline yielding $R^2$ = 52%. This indicates that higher interest rates coincide with lower levels of household income during the sample period. Similarly, the second scatterplot produced $R^2$ = 44%, suggesting a weaker negative relationship between RMHI and level of unemployment. This result is consistent with expectations regarding labor market conditions and household earnings.



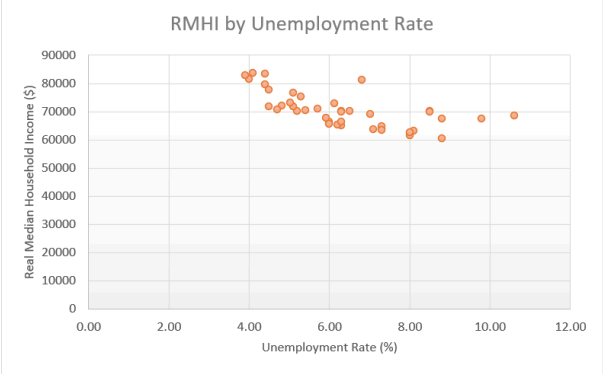




Figure 4: Real Median Household Income and Market Yield of 10-yr Treasuries
Figure 5: Real Median Household Income and Unemployment Rate

3. *EDA: Yahoo Finance (Equity) Dataset*

Summary statistics were also computed for the equity dataset within the sample of 40 companies (Table 2). These statistics were calculated using the sample daily dataset **prior to the monthly or yearly aggregation** step (n = 47,808). The mean opening and closing prices are *very* close ($119.95, $119.96) while the median values are closer to $93, indicating a right-skewed distribution of stock prices. This is expected, as a small number of pricey stocks increase the average relative to the median. The high and low price statistics display very similar distributions, with standard deviations near

$97 and ranges exceeding $650, reflecting the wide variation in price levels across different firms included in the sample.

| Open | |
|---|---|
| **Mean** | 119.9565 |
| Standard Error | 0.442616 |
| Median | 93.62617 |
| Mode | 140 |
| **Standard Deviation** | 96.77825 |
| Sample Variance | 9366.031 |
| Kurtosis | 5.049213 |
| Skewness | 1.893796 |
| Range | 660.3811 |
| Minimum | 2.32652 |
| Maximum | 662.7077 |
| Sum | 5734882 |
| Count | 47808 |

| High | |
|---|---|
| **Mean** | 121.5078 |
| Standard Error | 0.448242 |
| Median | 94.8 |
| Mode | 140 |
| **Standard Deviation** | 98.00837 |
| Sample Variance | 9605.641 |
| Kurtosis | 4.994494 |
| Skewness | 1.887248 |
| Range | 667.1241 |
| Minimum | 2.394947 |
| Maximum | 669.519 |
| Sum | 5809043 |
| Count | 47808 |

| Low | |
|---|---|
| **Mean** | 118.3817 |
| Standard Error | 0.436819 |
| Median | 92.38846 |
| Mode | 115 |
| **Standard Deviation** | 95.51056 |
| Sample Variance | 9122.268 |
| Kurtosis | 5.094536 |
| Skewness | 1.898997 |
| Range | 658.2594 |
| Minimum | 2.287418 |
| Maximum | 660.5468 |
| Sum | 5659590 |
| Count | 47808 |

| Close | |
|---|---|
| **Mean** | 119.9651 |
| Standard Error | 0.442574 |
| Median | 93.56073 |
| Mode | 133.8 |
| **Standard Deviation** | 96.76905 |
| Sample Variance | 9364.25 |
| Kurtosis | 5.044321 |
| Skewness | 1.892742 |
| Range | 662.1138 |
| Minimum | 2.326519 |
| Maximum | 664.4404 |
| Sum | 5735290 |
| Count | 47808 |

Table 2: Summary Statistics for Equity Prices (2018-2023)

Corporate actions (dividends and stock splits) occur far less frequently across the sample (Table 3). The median value for both variables is zero, indicating that most monthly observations contain no dividend payments or stock split events, given the large observation count of 47,808. However, occasional outliers produce high kurtosis and skewness. Trading volume shows *significant* variability, with an average monthly volume of 7.5 million shares but a maximum exceeding 914 million shares, highlighting differences in trading activity among firms.

| Volume | |
|---|---|
| **Mean** | 7533898 |
| Standard Error | 117829.1 |
| Median | 1989200 |
| Mode | 970400 |
| **Standard Deviation** | 25763374 |
| Sample Variance | 6.64E+14 |
| Kurtosis | 152.3632 |
| Skewness | 9.956448 |
| Range | 9.14E+08 |
| Minimum | 0 |
| Maximum | 9.14E+08 |
| Sum | 3.6E+11 |
| Count | 47808 |

| Dividends | |
|---|---|
| **Mean** | 0.00609 |
| Standard Error | 0.0003 |
| Median | 0 |
| Mode | 0 |
| **Standard Deviation** | 0.06551 |
| Sample Variance | 0.004292 |
| Kurtosis | 168.16 |
| Skewness | 12.41741 |
| Range | 1.55 |
| Minimum | 0 |
| Maximum | 1.55 |
| Sum | 291.1454 |
| Count | 47808 |

| Stock Splits | |
|---|---|
| **Mean** | 0.000353 |
| Standard Error | 0.000151 |
| Median | 0 |
| Mode | 0 |
| **Standard Deviation** | 0.033001 |
| Sample Variance | 0.001089 |
| Kurtosis | 14315.32 |
| Skewness | 112.7228 |
| Range | 5 |
| Minimum | 0 |
| Maximum | 5 |
| Sum | 16.875 |
| Count | 47808 |

Table 3: Summary Statistics for Volume and Corporate Actions (2018-2023)

Additional metrics were calculated after the dataset was aggregated monthly, including **monthly returns and rolling 12-month volatility** (Table 4). Monthly returns average approximately 1.18%, though the large standard deviation of 13.7% indicates high variability in stock performance across companies and time periods. Returns also display positive skewness and high kurtosis, suggesting occasional extreme price movements. Rolling 12-month volatility averages 10.26% with a standard deviation of 9.44% The difference in observation counts between the price statistics and these measures reflects the additional data requirements needed to compute these metrics, particularly the full 12-month window for volatility.

| Monthly Returns | |
|---|---|
| **Mean** | 1.180462 |
| Standard Error | 0.287033 |
| Median | 0.601257 |
| Mode | 0 |
| **Standard Deviation** | 13.69962 |
| Sample Variance | 187.6796 |
| Kurtosis | 10.71693 |
| Skewness | 0.512675 |
| Range | 200.3294 |
| Minimum | -85.5794 |
| Maximum | 114.75 |
| Sum | 2689.093 |
| Count | 2278 |

| Rolling 12-Month Volatility | |
|---|---|
| **Mean** | 10.2633 |
| Standard Error | 0.21793 |
| Median | 7.401364 |
| Mode | #N/A |
| **Standard Deviation** | 9.444194 |
| Sample Variance | 89.19279 |
| Kurtosis | 3.419924 |
| Skewness | 1.792779 |
| Range | 55.0395 |
| Minimum | 0.03548 |
| Maximum | 55.07498 |
| Sum | 19274.48 |
| Count | 1878 |

Table 4: Summary Statistics for Equity Returns and Volatility (2018-2023)

Figure 6 illustrates total monthly trading volume across all sampled stocks. Activity fluctuates substantially over the period, peaking in early 2020. This corresponds to the onset of the pandemic, when markets experienced *significant* volatility and uncertainty regarding future expectations. Trading volume declined throughout 2021, before gradually increasing in late 2022, perhaps with the beginning of the tech boom.

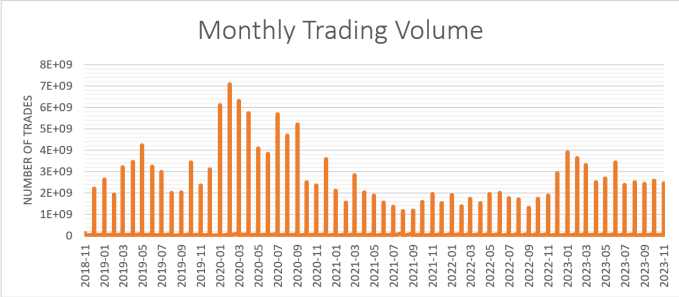
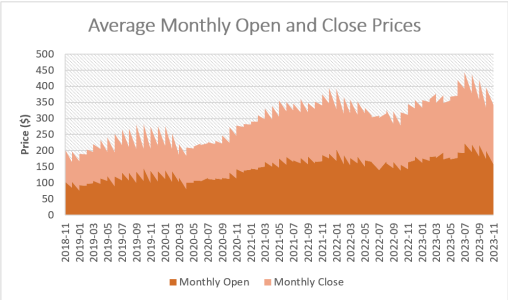



Figure 6: Total Monthly Trading Volume (2018-2023)

Figures 7 and 8 display various average stock pricing metrics across all sampled companies. Both graphs exhibit an upward trend in prices throughout the period, reflecting strong performance within equity markets. A substantial drop occurs during early 2020, consistent with the economic downturn associated with COVID-19. However, prices quickly recovered and continued to trend upward, experiencing only slight fluctuations throughout 2022 and 2023. These patterns illustrate how market activity and equity prices respond to *modern* periods of disruption.



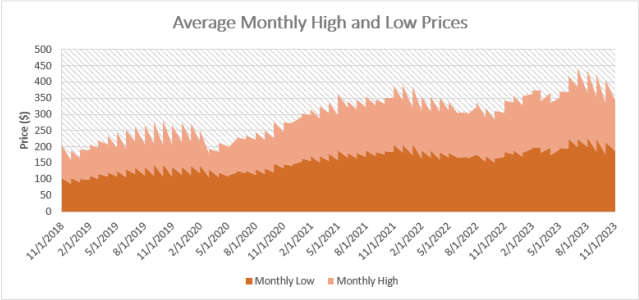
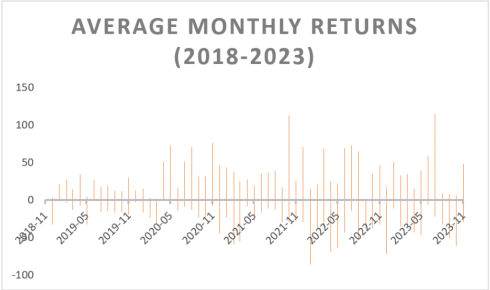




Figure 7: Average Monthly Open/Close Price (2018-2023)
Figure 8: Average Monthly High/Low Price (2018-2023)

Figures 9 and 10 plot the behavior of monthly equity returns across all sampled companies. Figure 9 presents the average monthly return, where returns stay moderate in the earlier portion of the sample period but begin to fluctuate more following COVID-19. Despite these fluctuations, returns are generally stable around a central tendency of **zero**. Figure 10 shows the distribution of monthly returns, appearing normally distributed. This result is consistent with financial modeling assumptions, where asset returns are often assumed to be normal.



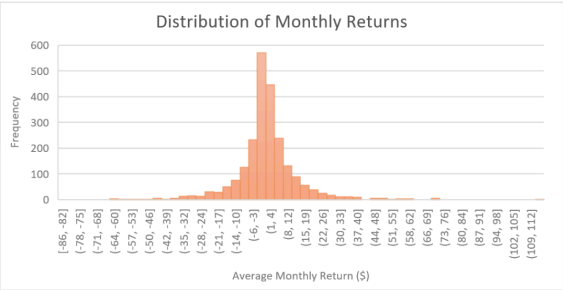
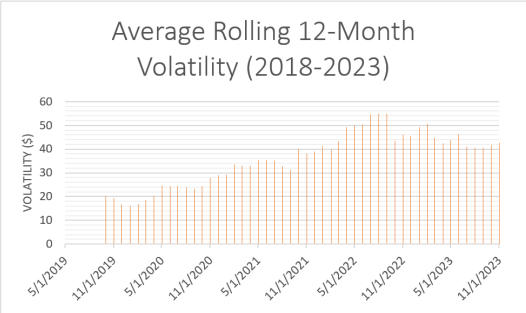




Figure 9: Average Monthly Equity Returns (2018-2023)
Figure 10: Distribution of Monthly Equity Returns (2018-2023)

Figures 11 and 12 examine volatility trends across all companies. Figure 11 displays the average rolling 12-month volatility and shows an upward trend until approximately November 2022, after which volatility declines before stabilizing at a high level. Figure 12 shows the distribution of rolling 12-month volatility, which is *heavily* right-skewed. The majority of observations center at lower volatility levels, with a small number of outliers exhibiting higher volatility. This demonstrates the importance of risk management measurements such as VaR, designed to estimate the **maximum potential loss** during these periods.



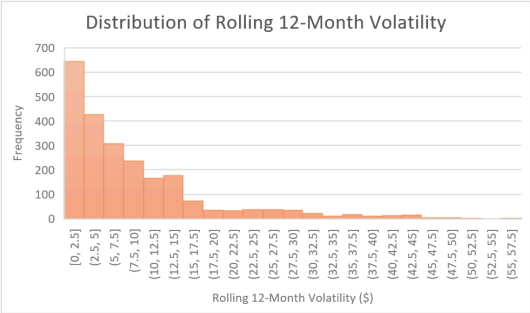




Figure 11: Average Rolling 12-Month Volatility (2018-2023)
Figure 12: Distribution of Rolling 12-Month Volatility (2018-2023)

4. *EDA: Company Analysis*

In addition to aggregated market behavior, several visualizations comparing trading activity across the sampled companies were created, highlighting differences in volume traded, corporate actions, and relative performance metrics. Figure 13 displays total trading volume across the entire sample period for each company, with results showing large variation in activity between firms. Firms like TSLA, BBD and ET exhibit *significantly* higher trading volume than others.

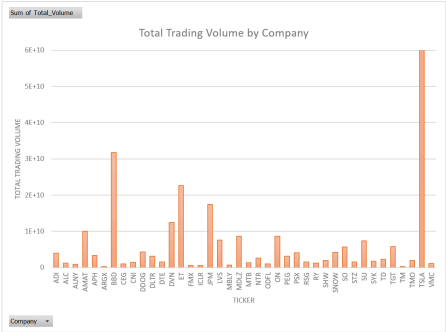
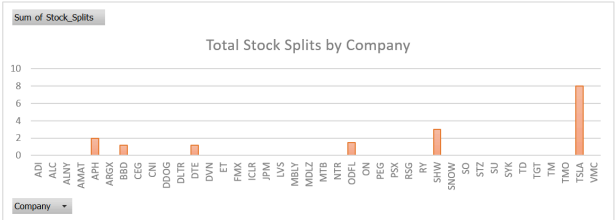



Figure 13: Total Trading Volume by Company (2018-2023)

Figures 14 and 15 reflect the total corporate actions taken across the sampled companies. Only six companies experienced stock splits, suggesting these firms (e.g. TSLA, SHW, and APH) may have experienced stronger price increases relative to others during the sample period. Figure 15 reveals the variation in dividend distributions among firms, with some companies providing substantial payments (DTE, JPM, MTC, PSX) and others paying none (ALNT, DLTR, TM).



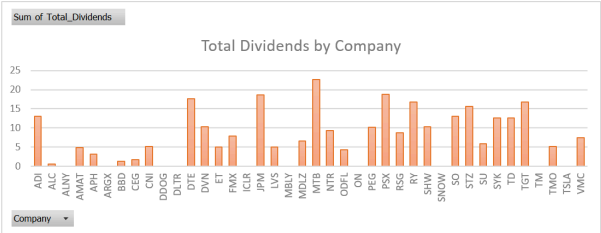
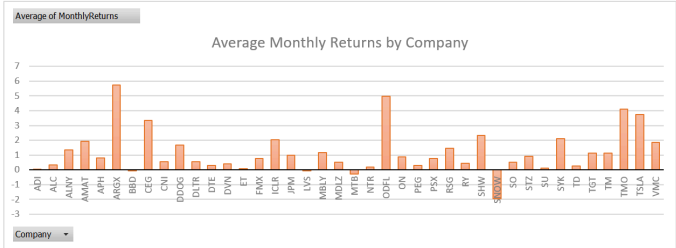




Figure 14: Total Stock Splits by Company (2018-2023)
Figure 15: Total Dividends by Company (2018-2023)

Figures 16 and 17 compare the average monthly returns and rolling 12-month volatility by company, respectively. While most firms exhibit small returns, some demonstrate high performance (ARGX, ODFL, TMO, TSLA) while others demonstrate *negative* returns (BBD, MTB, SNOW). The volatility results indicate differences in risk across the sample firms, with some companies experiencing significantly higher volatility (ARGX, SNOW, TMO, TSLA) than others (ADI, BBD).



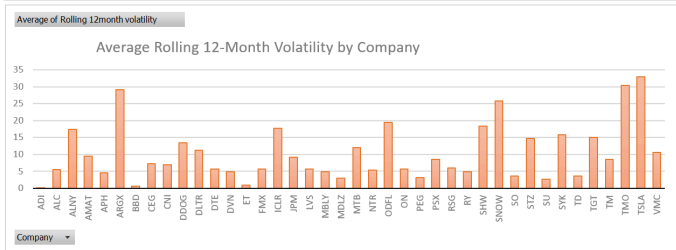
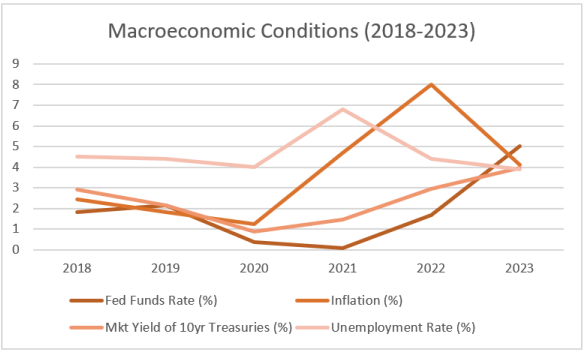




Figure 16: Average Monthly Equity Returns by Company (2018-2023)
Figure 17: Average Rolling 12-Month Volatility by Company (2018-2023)

## 5. *EDA: Macro-Equity Relationships*

Figure 18 graphs the same macroeconomic indicators as Figure 1 during the *equity* sample period from **2018-2023**. Rates reflect the economic downturn experienced as a result of the COVID-19 pandemic, followed by rapid increases throughout 2022. This is consistent with the tech boom following ChatGPT's release in November 2022. In 2023, rates converge to approximately 4%. Figure 19 extracts inflation and plots its time series along with the average monthly equity returns over the five-year sample period. The brief analysis demonstrates that inflation and average monthly returns tend to move in opposite directions.



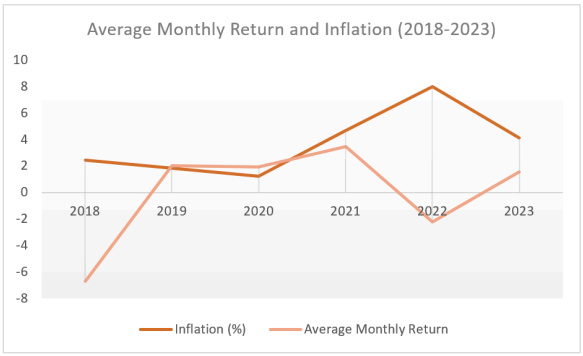
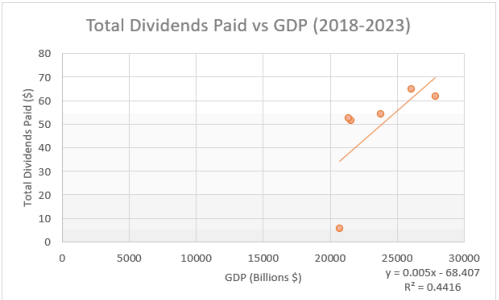
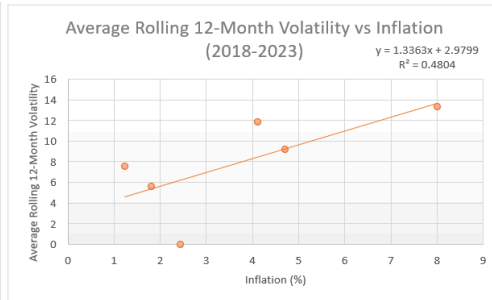



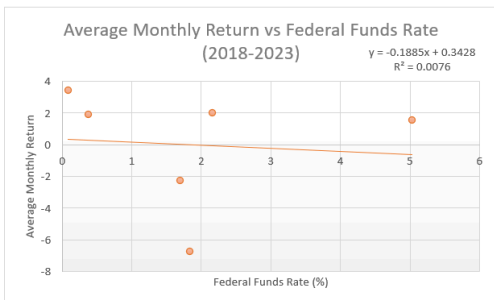

Figure 18: Macroeconomic Conditions (2018-2023)
Figure 19: Average Monthly Return and Inflation Rate (2018-2023)

Figures 20-23 attempt to estimate simple **OLS regression models** from the aggregated yearly data across both datasets. Figure 20 plots GDP against total dividends, demonstrating a positive relationship. With $R^2$ = 44.16%, this model exhibits the second strongest explanatory power. Figure 21 plots inflation against average rolling 12-month volatility; there is a similar linear association with $R^2$ = 48.04%. Figure 22 plots the federal funds rate against average monthly returns. There is a *very* weak, negative

correlation, with $R^2$ = 0.76%. Similarly, Figure 23 plots the unemployment rate against average monthly returns and experiences a weak positive association with $R^2$ = 8.96%.



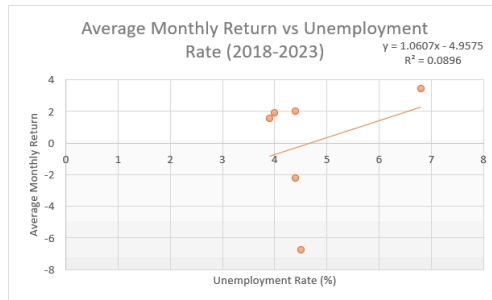




Figure 20: GDP (USD Billions) and Total Dividends (2018-2023)
Figure 21: Inflation and Average Rolling 12-Month Volatility (2018-2023)
Figure 22: Federal Funds Rate and Average Monthly Return (2018-2023)
Figure 23: Unemployment Rate and Average Monthly Return (2018-2023)

## *III. Methodology: Model Development and Application*

To predict equity returns, the majority of models selected were linear regression, with the exception of one Random Forest model for performance comparisons. When merging datasets for model development, one issue became clear: although the datasets had a 5-year overlap, the macroeconomic data was reported in *years*, while the equity data had *already* been aggregated monthly in a previous step. Once the sample equity data was aggregated yearly, there existed only 6 observations for comparison. Given that such a small sample size could lead to overfitting, **only one of the four models developed predicts returns *using* the macroeconomic factors as input variables**. This represents a significant limitation of my project, as the research hypothesis aimed to determine whether equity returns could reliably be predicted using macro inputs.

1. **Model 1: Linear Regression on Equity Dataset**
    a. **Dependent** (Y): Future Equity Monthly Returns
    b. **Independent** (X): Lag Returns, Lag Volatility, Lag Volume, and Lag Range

```
Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)     4.864e-01  6.935e-01   0.701  0.48320
Lag_Return      1.044e-01  3.208e-02   3.254  0.00118 **
Lag_Volatility -4.651e-02  9.748e-02  -0.477  0.63337
Lag_Volume      7.364e-12  7.957e-10   0.009  0.99262
Lag_Range       5.376e-02  4.604e-02   1.168  0.24324
---
```

The equity regression model demonstrates little explanatory power, with $R^2$ = 1.25%, indicating that a *very* small portion of variability in returns is captured by the included predictor variables (Lagged Returns, Volatility, Volume and Range). However, the overall model is statistically significant at the 0.05 alpha level (p = 0.0128), suggesting some predictive value across the *collective* variables. Among the predictors, only lagged returns are statistically significant (p = 0.00118), implying a present relationship between past and future returns.

2. **Model 2: Linear Regression on Macroeconomic Dataset**
    a. **Dependent** (Y): Inflation Rates
    b. **Independent** (X): Lag Federal Funds Rate, Lag GDP, Lag Unemployment Rate, Lag M2 Money Supply

```
Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)       4.2902532  2.7039049   1.587   0.1262
Lag_Interest     -0.0412898  0.1449373  -0.285   0.7783
Lag_GDP          -0.0004389  0.0002364  -1.857   0.0762 .
Lag_Unemployment  0.0189435  0.1973975   0.096   0.9244
Lag_M2            0.0005496  0.0002364   2.325   0.0292 *
---
```

The macro regression model demonstrates slightly stronger explanatory power, with $R^2$ = 26% and an adjusted $R^2$ = 13%. Once the model's complexity has been accounted for, the predictors explain only half has much of the variation in inflation. Among the predictors, Lagged M2 is the only statistically significant variable (p = 0.029), suggesting that money supply may have a positive predictive relationship with inflation. The overall model is **not statistically significant** (F-test p = 0.126), implying it lacks predictive capability. However, the model's RMSE = 1.41 suggests relatively little prediction error. While select macroeconomic indicators may influence inflation, the model as a whole is insufficient, likely due to potentially omitted variables.

3. **Model 3: Linear Regression on Combined Dataset**
    a. **Dependent** (Y): Future Equity Monthly Returns
    b. **Independent** (X): Lag Inflation Rates and Lag Federal Funds Rates

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)     1.3402     2.4284   0.552    0.636
Lag_Interest    0.9907     1.2834   0.772    0.521
Lag_Inflation  -0.3351     0.4331  -0.774    0.520

Residual standard error: 2.398 on 2 degrees of freedom
Multiple R-squared:  0.3642,    Adjusted R-squared:  -0.2716
F-statistic: 0.5728 on 2 and 2 DF,  p-value: 0.6358
```

The regression model merging **both** datasets demonstrates *very* weak reliability, primarily due to the miniscule sample size once datasets were combined. Although the model reports a moderate $R^2$ = 36%, the adjusted $R^2$ = -27%, indicating that the model can **not** explain variation in returns once the complexity of the model has been adjusted for. None of the predictors are significant at the 0.05 level, and the overall model is not significant (F-test p = 0.636). The large standard errors further suggest that the

estimates are **unstable and not reliable**. These results highlight the key limitation of combining the datasets: the lack of overlapping observations *severely* restricts the ability to build an informative model, rendering the results inconclusive.

4. **Model 4: Revised Equity Model: Random Forest**
    a. **Dependent** (Y): Future Equity Monthly Returns
    b. **Independent** (X): Lag Returns, Lag Volatility, Lag Volume, and Lag Range

```
               Type of random forest: regression
                     Number of trees: 100
No. of variables tried at each split: 2

          Mean of squared residuals: 251.8605
                    % Var explained: -16.6
```

The random forest regression model exhibits poor predictive performance, indicated by a *negative* percentage of variance explained (-16.6%). The RMSE = 17.45 demonstrates substantial prediction error compared to the scale of returns. Variable importance metrics show that Lagged Volatility, Lagged Range, and Lagged Return contribute *positively* to the model's predictive structure, while Lagged Volume provides little to no value and may introduce noise. This indicates that the additional complexity of the random forest does *not* translate into improved predictive accuracy and instead amplifies noise within the data. Overall, the regression model is more reliable and interpretable, whereas the random forest model is unsuitable in this context due to weak underlying relationships and the inherently noisy nature of financial return data.

5. **Model 5: Revised Macroeconomic Model: Linear Regression**
    a. **Dependent** (Y): Future Inflation Rates
    b. **Independent** (X): Change in Federal Funds Rate, Change in Unemployment Rate, Change in GDP (%), Change in M2 (%)

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.3483     0.5125   0.680    0.501
g_GDP           50.0770    10.0783   4.969 1.89e-05 ***
g_M2            -2.0032     4.7497  -0.422    0.676
d_Interest      -0.2605     0.1825  -1.428    0.162
d_Unemployment  -0.1568     0.2030  -0.773    0.445
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.117 on 34 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.4904,    Adjusted R-squared:  0.4304
F-statistic:  8.18 on 4 and 34 DF,  p-value: 9.844e-05
```

The revised macroeconomic regression model attempted to predict inflation rates based on the same input variables, but transformed in an effort to make results more meaningful. First, the input variables (Lagged GDP, Lagged M2, Lagged Federal Funds Rate, and Lagged Unemployment Rate) were transformed into *change* variables from the present and previous periods. For GDP and M2, these variables were further adjusted as percentage changes. Once predictors were adjusted, the model's explanatory power nearly doubled ($R^2$ = 49.04%, Adj. $R^2$ = 43.04%). The model is

significant overall with an F-test p = 9.844e-5, but the only significant *variable* at the 0.05 level is now the percentage change in GDP.

## *IV. Discussion*

*Comparing Model Performance: RMSE*

Across all 5 models developed, RMSE, or the average difference between actual and predicted values, was consistently **higher when predicting monthly equity returns** than macroeconomic factors. Models 1 and 4, which attempted to predict monthly equity returns based on Lagged Monthly Returns, Lagged Volatility, Lagged Volume and Lagged Price Range, had an average RMSE = 17.140394. This error is *significantly* higher than Models 2 and 5 (average RMSE = 1.22232), which attempted to predict Inflation Rates based on different combinations of macroeconomic variables.

| model | RMSE |
|---|---|
| 1 | 16.832375 |
| 2 | 1.408596 |
| 3 | 1.516798 |
| 4 | 17.448413 |
| 5 | 1.036044 |

*Project Changes*

Initially, the project aimed to answer whether various macroeconomic variables impacted stock returns and volatility over time. However, as the data was prepared for modeling, limitations forced the project's focus to shift towards analyzing each dataset individually. Since the equity dataset was reported daily and the macroeconomic dataset annually, the former needed to be aggregated monthly and then yearly. Once the equity data was aggregated yearly, there were only **six observations** to compare with the macroeconomic dataset (2018-2023), representing a clear limitation. Therefore, only one model attempts to predict equity returns using the merged dataset. The other models attempt to answer slightly different questions, such as:

- How do **shock periods** impact equity performance over time?
- Can inflation reliably be predicted by other macroeconomic variables?
- How does equity performance vary by **company** over the sample period?

*Conclusion*

The results of this analysis suggest that both equity returns and macroeconomic outcomes exhibit **limited predictability** when modeled using standard statistical techniques. Despite testing multiple models across both datasets with varying inputs, predictive performance remained weak and inconsistent.

For the equity models, the low explanatory power and high prediction error indicate that past market information such as returns, volatility, volume, and price range provide **minimal insight** into **future returns**. While lagged returns showed *some* statistical significance, this relationship was not strong enough to meaningfully improve overall model performance. The

failure of the random forest model to outperform linear regression further reinforces the idea that the underlying data lacks stable patterns. On the other hand, the macroeconomic models demonstrated relatively stronger performance, particularly after transforming variables into changes and growth rates. This suggests that macroeconomic relationships are more effectively captured when variables are expressed in terms of *movement* rather than levels. However, even the *improved* model was limited, with only a subset of variables contributing meaningfully, indicating that inflation dynamics are influenced by factors not captured in this analysis.

These findings highlight a key difference between **financial** and **macroeconomic** data. While macroeconomic variables may exhibit *some* degree of structured relationship, equity returns appear to be significantly more volatile and **difficult to predict**, aligning with financial theory. Ultimately, the results of this project do **not** support the hypothesis that macroeconomic variables can reliably predict equity returns within the constraints of the available data. That said, this outcome provides meaningful insight into the *real* challenges of financial modeling. The inability of multiple models to generate strong predictive power suggests that **noise, data limitations,** and the **complex nature of financial markets** contribute significantly to market performance.

*Future Research*

In a future iteration of the project, it would be helpful to obtain either macroeconomic data reported on a daily/monthly basis or annual stock data from a wider time period. Due to the time reporting differences across the two datasets, forming a decisive conclusion to support the original research hypothesis is **impossible**. With so few observations in the merged dataset, forming a model with significant results is highly prone to overfitting. To determine whether macroeconomic factors impact equity returns and volatility, **more observations are necessary**. Furthermore, to address the impact of shock periods on stock performance, a wider range of equity data should be collected, particularly one that captures the 2008 financial crisis, the 2000 dot-com boom, and the rising inflation throughout the 1970-80s.

In order to draw more definitive conclusions about the influence of macroeconomic factors on *different* financial instruments, it would also help to obtain a third dataset tracking bond or derivative performance metrics over a similar time period. Since equity returns are notoriously difficult to predict and model, perhaps models would perform better when predicting the performance of assets less impacted by daily noise.